# Comprehensive Hyper-Local Aviation Weather Verification & Multi-Model Fore

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**EXECUTIVE ABSTRACT & SYSTEM SCOPE**

Final comprehensive group thesis presenting the end-to-end design, data integration pipeline, machine learning architectures, and cloud-native verification platform for Bandaranaike International Airport (VCBI). Synthesizes individual contributions into an operational aviation support system.

## 1. Executive Project Summary

This thesis addresses the critical challenge of aviation safety under volatile Sri Lankan monsoon regimes. Integrating official observational archives from the Department of Meteorology with state-of-the-art AI, the project delivers a unified verification dashboard.

## 2. System Architecture & End-to-End Pipeline

The system connects live Aviation Weather API endpoints with four dedicated ML microservices:

1. Atmospheric State Engine (Temperature, QNH & Density Altitude)
2. Runway Aerodynamics Module (Wind speed/direction, Runway 04/22 crosswind vectors)
3. Visibility & Ceiling Classification Module (Fog, mist, low cloud threshold warnings)
4. Humidity & Pressure Trend Pipeline (Dew point, relative humidity derivation, barometric tendency)

## 3. Deployment, Security & Operational Verification

Implemented with a high-performance React/Vite front-end and a Python predictive core, the system features sub-second response times, automated METAR/TAF parity checks, and interactive visual analytics for pilots and dispatchers.